

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	
Team ID	
Project Name	Optimizing User, Group, and Role Management with Access Control and Workflows
Maximum Marks	4 Marks

Architecture Diagram:

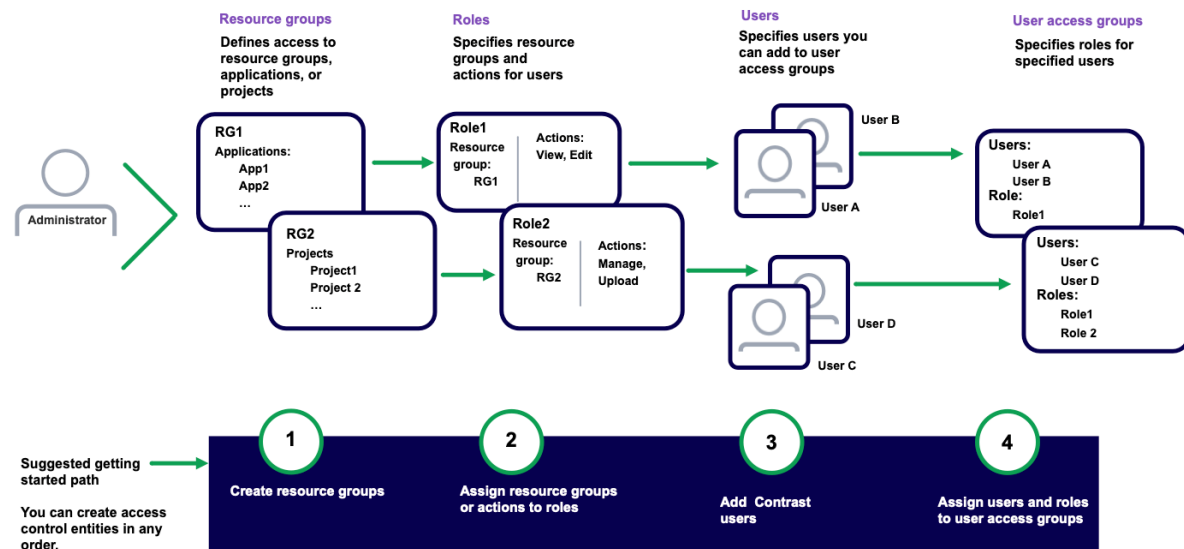


Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Provides a webbased interface for administrators and users to manage records, tasks, and permissions.	ServiceNow Web UI
2	User Management	Creates and manages user accounts with appropriate access rights.	ServiceNow User Administration

3	Group Management	Organizes users into groups for project collaboration and task assignment.	ServiceNow Groups
4	Role Management	Assigns roles such as Project Manager and Team Member to users.	ServiceNow Roles
5	Access Control	Restricts record and field access based on user roles using Access Control Lists.	ServiceNow ACLs
6	Workflow Automation	Automates approvals, notifications, and task assignments.	Flow Designer
7	Business Logic	Executes validation rules and business processes automatically.	Business Rules
8	Database	Stores users, groups, roles, projects, and task records securely.	ServiceNow Tables
9	Notification Service	Sends automated email notifications for approvals and task updates.	ServiceNow Notification Engine
10	Infrastructure	Hosts the complete application in a secure cloud environment with high availability.	ServiceNow Cloud Platform

Application Characteristics Security and Access Control

The system uses Role-Based Access Control (RBAC) and Access Control Lists (ACLs) to ensure that only authorized users can access specific modules and records. Project Managers have full administrative privileges, while Team Members have restricted permissions based on their assigned roles.

Workflow Automation

Business processes such as task assignment, approval requests, and email notifications are automated using ServiceNow Flow Designer. This minimizes manual effort and improves operational efficiency.

Data Management

All project-related information, including users, groups, roles, projects, and tasks, is stored securely within ServiceNow tables. The centralized data model ensures consistency and simplifies record management.

Cloud-Based Deployment

The application is hosted on the ServiceNow cloud platform, providing secure access, high availability, and reliable performance without requiring additional infrastructure.

Performance and Reliability

The platform efficiently manages user requests through optimized database operations, server-side processing, and built-in platform capabilities, ensuring smooth performance even as the number of users and records grows.

References

1. **ServiceNow Product Documentation** <https://docs.servicenow.com/>
2. **ServiceNow Developer Documentation** <https://developer.servicenow.com/>
3. **ServiceNow Product Documentation Portal** <https://www.servicenow.com/docs/>
4. **ServiceNow Community** <https://www.servicenow.com/community/>